

How do we get that

$$I_{\beta}(t) = \int_{\mathbb{R}^d} \frac{1}{(1 + |\xi|^2)^{1-\alpha}} \left(1 - e^{-t(|\xi|^2 + \beta)}\right) \hat{f}(d\xi)$$

When I use mathematica, I get that

$$I_{\beta}(t) = \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{(1 + |\xi|^2)^{1-\alpha}} \left[\Gamma(1 - \alpha) - \Gamma(1 - \alpha, t(\beta + |\xi|^2)) \right].$$

I used the following in mathematica to find the dr integral:

`Integrate[r^(-a) * Exp[-r *( z^2 + b)], {r,0,t}, Assumptions-> a>0 && z>0 && b>0 && t>0]`

Le's answer: What I have done is let $|\xi| = 0$ and ∞ to obtain the lower and upper bounds, namely,

$$\left(1 - e^{-\beta t}\right) \Upsilon_{\alpha}(\beta) \leq I_{\beta}(t) \leq \Upsilon_{\alpha}(\beta)$$

Please let me know if this is correct?